

Evan Keyu Xiang

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Education

- Massachusetts Institute of Technology**, Cambridge, MA (GPA: TBD/5.0)
Bachelor of Science in Electrical Engineering with Computing & Biology;
– **Selected Coursework:** TBD
- Shady Side Academy Senior School**, Pittsburgh, PA (GPA: 4.06/4.33)
High School Diploma & Cum Laude Society Inductee;
– SAT (1560, 760R, 800M) & PSAT (1520, 760R, 760M)
– Bausch & Lomb Honorary Science Award, Miller Award, National Merit Finalist

Experiences & Leadership

- Effluent - Founding ML Engineer** 2024 - 2026
• Developed and implemented custom graph models with dynamic thresholding to identify wastewater infrastructure anomalies. Backed by SV Angel, Susa Ventures, Pareto Ventures, etc.
- Research Intern @ Chang Lab (Stem Cell and Neuroregeneration Laboratory)** 2024 - 2026
• Conducted prospective study investigating chromatin compartment rearrangements near EMT-critical genes, identifying targets for genetic therapies to combat post-surgical cataracts.
- Trained Scribe - Shady Side Academy Writing Center** 2024 - 2026
• Competitively selected as 1 of 12 students by the English Department; oriented freshmen to the intro literature curriculum and supported faculty in teaching foundational academic writing.
- Science Olympiad Captain/Exam Writer** 2021 - 2026
• Led Division B to their best national placement in 18 years and advanced Division C eight places at the State Tournament; managed logistics for 7 overnight competitions per season.
• Invited as a test writer for 35 tournaments, including the NJ and NM State Tournaments; authored 39 tests across various fields, reaching 13,000+ students.
- Summer Science Program (SSP) in Genomics (University of Guelph)** 2025 - 2025
• Conducted adaptive laboratory evolution experiments with *V. natriegens* under spectinomycin stress to identify mutations conferring antibiotic resistance; designed poster & presented to UoG faculty.
- Research Intern @ Treble-Barna Traumatic Brain Injury Lab** 2024 - 2025
• Conducted research linking brain-derived neurotrophic factor (BDNF) levels to post-TBI behavioral outcomes in pediatric populations via K-means cluster analysis; supported end-to-end sample processing.

Selected Awards & Honors

- **USA Biology Olympiad National Finals - Silver Medalist** 2025
Placed Top 20 among biology students in the country twice & invited to National Finals. Placed Top 8 at the 2025 USABO National Finals at Harvard. Recognized by Sen. Fetterman & McCormick.
- **Regeneron ISEF Grand Prize Winner - 3rd in Biomedical and Health Sciences** 2025
Earned a grand award for studying mechanisms of treatment and molecular pathways underlying posterior capsular opacification, including novel Cas9-based therapies targeting chromatin architecture.
- **USA Earth Science Olympiad National Finals Invitee** 2024
Invited as one of the finalists for the USA Earth Science Team, unable to attend due to conflict.
- **Pennsylvania Junior Academy of Science (PJAS) - 1st Prize in Biology** 2024
2025 - Regional/State 1st Place + Perfect Score; 2024 - Regional 1st Place
- **Science Olympiad National Medalist** 2024
National Runner Up in Trebuchet and 4th place in Meteorology at the 2023 National Tournament, held at Wichita State University. Led team to first top 10 national finish in 15 years.

Projects

Xiang, E., Wang, T., & Poddar, V. (2024, December 2). *High-Throughput Detection of Risk Factors to Sudden Cardiac Arrest in Youth Athletes: a Smartwatch-Based Screening Platform*. arXiv.org. <https://arxiv.org/abs/2412.12118>

Technical Skills

- **Statistics & Programming:** Python, MatLab, R, SPSS Statistics v28, L^AT_EX, Graphpad PRISM (Figures), C++
– **Lab Skills:** Western Blot, Mammalian Cell Culture, RT-qPCR, ChIP-qPCR, Sequencing (Library Prep for Hi-C/NGS), etc.